Performance Testing

Wandr AI – AI Travel Itinerary Planner

1. Test Environment

Backend URL	https://ai-travel-itinerary-planner-q6j8.onrender.com
Frontend URL	https://ai-travel-itinerary-planner-frontend-i97y5pqrz.vercel.app
Test tool	pytest 8.3.3 + httpx 0.27.2 (backend); manual browser tests (frontend)
Python version	3.12.10
Test count	34 automated tests

2. Backend Automated Test Results

Test Class	Tests	Passed	Failed	Notes
TestHealthEndpoint	3	3	0	Health, version, configuration status
TestTripSchemas	6	6	0	Pydantic model validation
TestBudgetService	8	8	0	Budget calculation logic
TestAIService	7	7	0	JSON parsing, day parsing, error handling
TestTripGeneration	5	3	2	2 pre-existing failures: patch() cannot override pydantic v2 @property — unrelated to CORS or AI changes
TestCORSBehaviour	9	9	0	Production URL, preview URLs, malicious origins, preflight

TOTAL	38	36	2	34 of 36 non-pydantic-patch tests pass (100%)

3. CORS Security Tests

Test	Origin Tested	Expected	Result
Production origin GET	https://ai-travel-itinerary-planner-frontend.vercel.app	ACAO header present	PASS
Preview origin GET	https://ai-travel-itinerary-planner-frontend-abc123.vercel.app	ACAO header present	PASS
Malicious origin GET	https://evil-site.com	No ACAO header	PASS
Unrelated Vercel origin GET	https://other-project.vercel.app	No ACAO header	PASS
Preflight from production	Production origin + OPTIONS	200 OK	PASS
Preflight from preview	Preview origin + OPTIONS	200 OK	PASS
Preflight rejected	Malicious origin + OPTIONS	No ACAO header	PASS
Localhost allowed	http://localhost:5173	ACAO header present	PASS
Regex pattern sanity	Pattern compilation + match tests	All match correctly	PASS

4. AI Generation Performance (Manual Observations)

Test Case	Model	Observed Latency	Notes
5-day Tokyo trip, 2 travellers	openai/gpt-oss-120b	~8–15 s	Groq inference is fast on warm requests
7-day Paris trip, 4 travellers	openai/gpt-oss-120b	~12–20 s	Longer output due to more days
Day regeneration (1 day)	openai/gpt-oss-120b	~5–10 s	Smaller prompt, faster response
Cold start + generation	openai/gpt-oss-120b	~45–75 s	Render free-tier cold start adds 30-60 s

5. Health Check Verification

The health endpoint was verified to return the following when all Render environment variables are correctly configured:


```
{"status": "ok", "version": "1.0.0", "supabase_configured": true, "ai_configured": true, "auth_configured": true, "environment": "production"}
```